Newell Dodoh

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EDUCATION

Indiana University Bloomington – Luddy School of Informatics, Computing, Engineering **May 2028**
Bachelor of Science in Computer Science, Data Science & Machine Learning Concentration GPA 3.73/4.0

- Minors: Business & Psychology
- Awards: Hudson and Holland Scholar, Dean's List (3 Terms), Outstanding Research in Technology & Computing Award

Coursework: Cloud Computing, Data Structures, Discrete Structures, Statistics, Software Systems, Computer Networks, Data Mining, Mobile App Development, Applied Algorithms

TECHNICAL SKILLS

Programming: Python, Java, C, C++, C#, SQL, JavaScript, HTML/CSS, Kotlin
Software Engineering: REST APIs, SQL Databases, Machine Learning, Software Development, Debugging, Deployment, Automation
Tools: GitHub, IntelliJ IDEA, Android Studio, JIRA, SharePoint, Microsoft Power BI, Tableau, Figma, Microsoft Office, Microsoft Copilot

EXPERIENCE

Kelley School of Business, Data Science and AI Lab | Undergraduate Research Intern August 2025 - Present
Bloomington, IN

- Fine-tuned Mistral-7B for mental health dialogue using supervised fine-tuning (SFT) and LoRA, a parameter-efficient fine-tuning method, raising the model's empathy score from 3.18 to 3.54.
- Designed multilingual LLM evaluations for four query-rewriting strategies across English and Mandarin Chinese, identifying culturally tailored Mandarin rewriting as the top-performing variant for empathy.
- Benchmarked six candidate LLMs to select Mistral-7B for multilingual adaptation and curated 1,250 dialogue pairs per dataset for training and evaluation.

Indiana University's President's Office | 105th Board of Aeons April 2026 - Present
Bloomington, IN

- Coordinate research projects with the Office of the President, helping define project scope, assess the feasibility of proposed solutions, and develop recommendations for a campus of 49,000+ students.
- Nominated and Selected by IU President Pamela Whitten as 1 of 12 members from 300+ applicants for the Board of Aeons, President Office's research and advisory board

Indiana University | Undergraduate Research Assistant June 2026 - July 2026
Bloomington, IN

- Developed a methodology for benchmarking LLM responses to folk neuroscience concepts against peer-reviewed scientific evidence
- Compared the epistemic reliability of leading LLMs when responding to questions about dopamine, addiction, and digital well-being
- Identified and adapted an open-source AI video summarization tool, updating a legacy code and dependencies to successfully deploy the application in a current computing environment

Deloitte & EY | Technology Consulting Awareness Readiness Program January 2026 - March 2026
Bloomington, IN

- Selected for a competitive 3-month program bridging technical implementation with business strategy through case studies
- Applied foundational consulting frameworks, including MECE, Issue Trees, and Porter's Five Forces, to decompose complex technical challenges into structured, manageable workstreams
- Applied Agile and Waterfall project management principles to simulate the deployment lifecycle of enterprise-level technology

PROJECTS

Spotify Music Genre Classification | Python, scikit-learn, pandas, Mathplotlib April 2026 - May 2026

- Built a 114-class Random Forest pipeline on 114,000 Spotify tracks, achieving 32.7% cross-validated accuracy - 37× above the random-guessing baseline
- Designed a full Machine Learning evaluation pipeline including stratified 80/20 splits, 5-fold cross-validation, per-class precision/recall/F1 scoring, and confusion matrix analysis across 114 genre categories

JPMorganChase Software Engineering Job Simulation on Forage March 2026

- Implemented transaction validation and persistence logic with Spring Data JPA and an SQL database
- Developed a REST endpoint for querying user balances, returning JSON responses through a Spring controller

Women's Research Hackathon | Team Lead | Bloomington, IN February 2026

- Led a hackathon team from defining project scope to delivering a working prototype that tracks women's representation in UN dataset
- Translated findings from a United Nations dataset into product priorities, using global representation trends to determine what the prototype should track.

LEADERSHIP & COMMUNITY INVOLVEMENTS

Alpha Kappa Alpha Sorority, Incorporated, Tau Chapter | Vice President | Bloomington, IN Nov 2025 - Present

- Appointed by International CEO & President to serve on the International Technology Committee, impacting 390,000+ global members
- Appointed by Regional Director to serve as an Undergraduate Cluster Coordinator, planning conferences for attendees from 8 states
- Selected as a *2026 International Leadership Fellow*, the Sorority's premier and prestigious leadership development initiative

National Society Of Black Engineers, Indiana University Chapter | Vice President | Bloomington, IN April 2025 - Present

- Oversee operational planning and execution for an engineering-focused student organization with 40+ paid members
- Partner with industry-facing companies to support programming aligned with engineering, technology, and career advancement